🍽️ BIOLOGY CH.13 — DIGESTIVE SYSTEM — FULL MIND MAP

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📑 Topics in this chapter:

1. Bile & Fat Emulsification

- Bile salts large fat globules ko chhote globules mein todte hain (emulsification) — 'soap on dirt' jaisa kaam karte hain, jisse enzymes zyada effectively work kar sakein. *[Q728, Q732, Q746, Q750]*
- ⚠️ Trap: Bile juice mein protein-digesting enzyme NAHI hota — bile fat globules break karti hai, digest nahi karti. Bile juice: germs kill karti hai, fat absorption help karti hai, chyme ka acid content kam karti hai, pancreatic enzymes ke liye alkaline medium deti hai. *[Q744]*
- Bile juice 'alkaline medium' deti hai — stomach se aane wale acidic chyme ko neutralize karti hai, pancreatic enzymes work karne ke liye suitable environment banati hai. *[Q756]*
- Bile 'Gallbladder' mein store-concentrate hoti hai — liver dwara produce hoti hai, bile salts-phospholipids-cholesterol-bilirubin-electrolytes-water se bani hai. *[Q765]*
- Bile juice 'Liver' se secrete hoti hai, gallbladder mein store hoti hai — fat digestion-absorption mein 2 important functions perform karti hai. *[Q774]*
- Digestion mein bile ka role hai 'fat ka emulsification' — bile acids large lipid droplets ko chhote mein todte hain, surface area badhate hain digestive enzymes ke liye. *[Q775]*

2. Stomach — HCl & Gastric Juice

- ⚠️ Trap: Human stomach 'nitric acid' NAHI, 'Hydrochloric acid' (HCl) produce karta hai — HCl food digest karne mein help karta hai, Vitamin B12 absorption ke liye zaroori hai. *[Q729, Q777]*
- 'Pepsin' protein-splitting enzyme hai — inactive pepsinogen se HCl ki presence mein banta hai. Ptyalin/Amylase carbohydrates, Lipase fats digest karta hai. *[Q730]*
- Pepsin ko digestion mein action ke liye HCl (Hydrochloric acid) ki presence chahiye — chief cells inactive pepsinogen secrete karti hain, Parietal cells HCl secrete karti hain jo pH kam karke pepsinogen ko activate karti hai. *[Q739]*

- 'Mucus' stomach ki inner lining ko uske apne acid se protect karta hai — gastric glands gastric juice (HCl + Pepsin + Mucus) secrete karti hain. *[Q760]*
- Stomach ka shape English alphabet 'J' jaisa hai — digestive tube ka sabse wide part hai, food ko store-churn-mix karta hai gastric juices ke saath. *[Q761]*
- ⚠ Trap: 'Stomach liver-pancreas ki secretions receive karta hai' — ye statement INCORRECT hai. Actually SMALL INTESTINE (duodenum) liver (bile) aur pancreas (pancreatic juice) ki secretions receive karta hai, stomach NAHI. *[Q763]*
- Acid secretion stomach ki characteristic hai — Gastric acid (HCl) Parietal/oxyntic cells se secrete hoti hai, pepsinogen activate karti hai aur bacteria inactivate karti hai. *[Q771]*

3. Small Intestine — Structure, Length & Digestion Completion

- Small intestine 'Herbivores' (jaise Cow) mein Carnivores se lambi hoti hai — cellulose (grass mein) digest karne mein zyada time lagta hai. *[Q731, Q773, Q781]*
- Villi ki characteristics jo absorption enable karti hain: (i) finger-like structures with thin walls, (ii) large surface area provide karte hain, aur blood vessels ka network close hota hai — villi mein pores absorption ke liye NAHI hote. *[Q734, Q747]*
- Herbivores jaise ghode-khargosh mein digested food ka cellulose 'Caecum' (large intestine ka pehla part) mein bacteria ki action se digest hota hai. *[Q751]*
- Proteins ka amino acids mein, carbohydrates ka glucose mein, aur fats ka fatty acids mein final conversion 'Intestinal Juice' se hota hai — intestinal glands se secrete hoti hai (maltase, lactase, lipase, Erepsin). *[Q752]*
- ⚠ Trap: 'Assimilation, ingestion se pehle aur absorption ke baad aata hai' — ye statement FALSE hai. Sahi order: Ingestion → Digestion → Absorption → Assimilation → Egestion. *[Q759]*
- Small intestine average adult human mein ~23 feet (7.5 meters) lambi hoti hai, highly coiled — liver-pancreas se secretions leti hai, wall bhi juices secrete karti hai. *[Q767]*
- ⚠ Trap: 'Alveolar sac' human alimentary canal ka part NAHI hai — ye LUNGS (respiratory system) ka part hai. Alimentary canal mein Mouth, Oral cavity, Pharynx, Oesophagus, Stomach, Intestine, Anus shaamil hain. *[Q776]*
- Carbohydrates, proteins aur fats ka complete digestion 'Small Intestine' mein hota hai — carbs glucose/fructose mein, proteins amino acids mein, fats fatty acids-glycerol mein todte hain. *[Q784]*

4. Salivary Enzymes & Mouth Digestion

- 'Amylase' saliva mein present digestive enzyme hai — salivary glands se produce hoti hai, starch aur fats digest karti hai. Enterokinase intestinal protein digestion enzyme hai, Peptidases peptide bonds cleave karti hain. *[Q735]*
- Rice, potatoes, bread mein 'Carbohydrates' hoti hain — ye breakdown hokar 'Glucose' banti hain. Salivary amylase mouth mein starch ko maltose mein todta hai, phir disaccharides monosaccharides mein. *[Q738]*

- 'Buccal cavity' (oral cavity) human 'Digestion' system ka part hai — mastication (teeth se grinding) se mechanical digestion hoti hai. *[Q742]*
- Salivary amylase (ptyalin) 'Starch' ko digest karne mein help karti hai — starch-glycogen ko glucose-maltose mein todti hai, pancreatic juice aur saliva dono mein present hoti hai. *[Q748]*
- 'Salivary amylase' (Ptyalin) starch ko simple sugar mein todne wala enzyme hai — 3 pairs of salivary glands se produce hoti hai: Parotids, Sub-maxillary, Sublinguals. *[Q749]*

5. Areolar Tissue

- 'Areolar tissue' — organs ke voids fill karta hai, internal organs ko support karta hai, aur tissue repair mein help karta hai — Adipose tissue fat store karta hai, Epithelial tissue surfaces cover karta hai. *[Q733]*

6. Lymphatic System — Fat Transport

- Lymph ke functions: digested fat ko intestine se carry karna, aur excess fluid extracellular space se blood mein drain karna — antibodies-lymphocytes bhi blood tak transport karta hai. *[Q737]*
- Lymph intestinal 'Fats' transport karta hai (specifically absorbed fats) — colorless fluid jo lymphatic system mein circulate karta hai. *[Q743]*

7. Dental Health & Teeth Structure

- Dental caries se 'soft enamel' hota hai — Enamel outer covering hai, human body ka hardest substance. Dentine tissue hai crown se root tak, Tooth pulp mein blood vessels-nerves hote hain. *[Q745]*
- Adult human mein 8 'Incisors' hote hain (4 upper + 4 lower) — total 32 teeth: 8 incisors, 4 canines, 8 premolars, 12 molars (4 wisdom teeth samet). *[Q757]*
- 'Teeth' calcium se bane hote hain — Calcium phosphate ($Ca3(PO4)2$) bones-milk-teeth mein milta hai, Enamel bones se bhi harder hota hai. *[Q768]*

8. Smooth Muscle Organs

- Smooth muscles wale organs: 'Iris of the eye, bronchi of lungs, ureters' — hollow visceral organs (liver, pancreas, intestines) mein bhi hoti hain (heart ko chhodkar). Bones mein smooth muscles NAHI hoti. *[Q740]*

9. Antacids — Indigestion Remedies

- Indigestion ke dauran zyada acid stomach mein produce hota hai — iski pain dur karne ke liye log 'bases' (antacids) use karte hain, jaise Milk of Magnesia. *[Q741, Q754, Q780]*

10. Ruminant & External Digestion

- 'Rumen' stomach ka ek chamber hai jo food store karta hai — cows, goats, sheep, deer mein milta hai, microbes feed components ko usable energy-protein mein convert karte hain. *[Q755, Q762]*
- 'Mushroom' body ke bahar food material breakdown karke digested food absorb karta hai — saprotrophic nutrition. Amoeba holozoic nutrition dikhata hai (ingestion-digestion-absorption-assimilation-egestion). *[Q758]*

11. The Pancreas — Islets of Langerhans

- 'Islets of Langerhans' Pancreas mein milte hain — endocrine (hormone-producing) cells ke regions hain, insulin-glucagon jaise hormones banate hain. *[Q764]*
- 'Pancreas' blood mein sugar content regulate karta hai — endocrine function (insulin-glucagon) aur exocrine function (digestive enzymes) dono hoti hai. *[Q766]*

12. Liver Functions & Digestive System Parts

- Liver, toxic 'Ammonia' ko 'Urea' mein convert karta hai (urea cycle) — bile juice bhi secrete karta hai (bilirubin, biliverdin, cholesterol), jo gallbladder mein store hoti hai. *[Q769]*
- ⚠ Trap: 'Heart' digestive system ka part NAHI hai — ye CIRCULATORY system ka part hai. Digestive system mein mouth, buccal cavity, pharynx, oesophagus, stomach, intestines, rectum, anus shaamil hain. *[Q770]*
- Liver ka function 'digestion' hai — bile secrete karke fats ka emulsification karta hai. Body ki largest gland hai (1.2-1.5 kg), abdominal cavity mein diaphragm ke neeche, 2 lobes. *[Q772, Q778]*

13. Energy Storage & Release

- Plants mein carbohydrates jo immediately use nahi hoti, unhe 'Starch' ki form mein store kiya jaata hai — potatoes, carrots, beets jaise plants roots-stems mein store karte hain. *[Q779]*
- Food se derived energy ka kuch hissa 'Glycogen' ki form mein store hota hai — polysaccharide compound hai, muscles-liver mein store hokar zaroorat par energy mein convert hota hai. *[Q782]*
- Digestion ke end mein release hone wali energy 'Chemical energy' ki form mein hoti hai — molecules ke chemical bonds mein stored energy. *[Q783]*

14. Match Lists — Digestive System Diagram & Functions

- Match: Trachea → no digestion; Villi → absorb nutrients; Small intestine → length depends on food organism eats; Liver → helps emulsification (fats). *[Q736]*
- Match: Mouth → intake of whole food; Teeth → chewing of food; Tongue → rolling of food; Saliva → swallowing of food. *[Q753]*